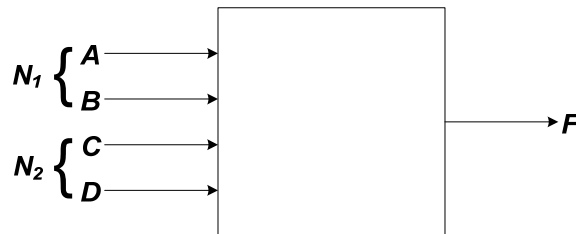


EE2020/EE2026 (Part 1)

Tutorial 4 - Questions

Boolean algebra and minimization

1. A switching circuit has four inputs as shown. A and B represent the first and second bits of a binary number N_1 . C and D represent the first and second bits of a binary number N_2 . The output is to be 1 only if the product $N_1 \times N_2$ is less than or equal to 2.
 - (a) Write the truth table for the system.
 - (b) Write the canonical SOP and POS expressions for F .
 - (c) Draw a Karnaugh-map (K-map) for the function F .
 - (d) From the K-map, derive a simplified SOP expression for F . (Hint: Use groups of only size 4.)



2. A bank vault has three locks with a different key for each lock. Each key is owned by a different person. To open the door, at least two people must insert their keys into the assigned locks. The signal lines A , B and C are 1 if there is a key inserted into lock 1, 2 or 3, respectively. Write an equation for the variable Z which is 1 if and only if the door should open.
 - (a) Write the truth table for the system.
 - (b) Write the canonical SOP and POS expressions for Z .
 - (c) Draw a Karnaugh-map (K-map) for the function Z .
 - (d) From the K-map, derive a simplified SOP expression for Z .
3. Use K-maps to obtain an MSOP and an MPOS for each of the following functions:
 - (a) $Z = \overline{A}\overline{B}\overline{C}D + \overline{A}B\overline{C}D + A\overline{B}\overline{C}D + A\overline{B}CD + A\overline{B}C\overline{D}$ with don't care for $ABCD = 1010$
 - (b) $Z = (\overline{A} + B + \overline{C})(A + B + \overline{C})$ with don't cares for $ABC = 111$ and 110
 - (c) $f(x_1, \dots, x_4) = \sum m(0, 4, 5, 6, 7) + D(1, 12, 13, 14, 15)$, where $m(\)$ is the set of minterms for which $f = 1$ and $D(\)$ is the set of don't cares. For example, $m(2)$ is the minterm corresponding to $x_1x_2x_3x_4 = 0010$ (this alternate shorthand notation is often used to express SOPs).
4. A combinational circuit has four inputs A , B , C and D and an output Z . The output is asserted whenever three or more of the inputs are asserted, otherwise the output is de-asserted. Find an MSOP expression for Z . Design combinational circuits using only 74'04 inverters, 74'00 2-input NAND gates and 74'10 3-input NAND

gates. Assume that A, B and Z are active high signals, while C and D are active low signals. Use alternate gate representations for clarity of circuit diagrams.